

A Component Exchange Is Outer: A Disproof of Conjecture 00000003380

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Abstract

The noncommutative complex algebra $M_2(\mathbb{C}) \times M_2(\mathbb{C})$ has an automorphism exchanging its two components. This automorphism moves a central idempotent, whereas every inner automorphism fixes the center pointwise. It is therefore outer, even though it preserves the usual matrix adjoint. The complete counterexample is verified in Lean.

The first explicit assertion in both languages of the source is that every automorphism of a noncommutative algebra is inner. There is no simplicity or factor hypothesis. We refute this universal assertion. An inner automorphism of a unital algebra is a map $a \mapsto uau^{-1}$ for a unit u of that algebra.

Theorem 1. *Let $\mathcal{A} = M_2(\mathbb{C}) \times M_2(\mathbb{C})$, with componentwise algebra operations. The map*

$$\sigma : \mathcal{A} \longrightarrow \mathcal{A}, \quad \sigma(X, Y) = (Y, X),$$

is a complex algebra automorphism, preserves the componentwise adjoint, and is not inner. The algebra $\mathcal{A}$ is noncommutative.

Proof. Component exchange preserves addition, multiplication, the unit (I, I) , and complex scalar multiplication. It is its own inverse. Moreover, $\sigma(X^*, Y^*) = (Y^*, X^*) = \sigma(X, Y)^*$.

The element $e = (I, 0)$ is a central idempotent: $e^2 = e$ and, for every $(X, Y) \in \mathcal{A}$,

$$e(X, Y) = (X, 0) = (X, Y)e.$$

Consequently every unit $u \in \mathcal{A}$ satisfies

$$ueu^{-1} = euu^{-1} = e.$$

But $\sigma(e) = (0, I) \neq (I, 0) = e$, so no unit of $\mathcal{A}$ can implement σ by conjugation. In particular no unitary of $\mathcal{A}$ can do so.

For noncommutativity, take

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $PQ = \text{diag}(1, 0)$ and $QP = \text{diag}(0, 1)$ differ. Hence $(P, 0)$ and $(Q, 0)$ do not commute in $\mathcal{A}$. □

The implementing unit must belong to the algebra whose automorphism is being considered. A permutation matrix in a larger ambient matrix algebra would not establish innerness in $\mathcal{A}$.

Formal verification and scope

The file `lean/Main.lean` uses actual 2×2 complex matrices and their product algebra. The object `exchange` has type $\mathcal{A} \simeq_{\mathbb{C}} \mathcal{A}$: it is a complete algebra equivalence, constructed from the library’s product ring equivalence and the checked scalar compatibility. The standard matrix conjugate transpose defines the componentwise adjoint; its preservation is proved explicitly.

The predicate `IsInner` quantifies over actual algebra units and all elements, with the formula $\varphi(a) = uau^{-1}$. Lean checks the central idempotent, its movement by exchange, its invariance under *every* such conjugation, and the actual products witnessing noncommutativity. The theorem `conjecture_false` directly negates the universal assertion that every complex algebra with a noncommuting pair has only inner complex algebra automorphisms.

The proof does not assume the counterexample is outer, substitute an unrelated numerical invariant, or weaken innerness to conjugation by particular matrices. It proves failure even when all invertible implementers are allowed, and thus covers the more restrictive unitary meaning as well. It does not dispute innerness results that impose additional hypotheses such as a central simple algebra. No conclusion about the source’s unspecified “measure of innerness” is needed: its first asserted universal clause already fails.

Source and reproduction

The exact bilingual source is included byte-for-byte in `SOURCE.md`; the original repository path is `conjectures/00000003380.md`. The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, followed by

```
lake env lean -DwarningAsError=true Main.lean.
```

The public Git dependency revisions are locked in the included manifest. Actual fresh-build logs, theorem-axiom reports, artifact hashes and PDF verification records are included in `verification/`. No numerical auxiliary experiment is required: the exact matrix and all-unit quantifier arguments are proved directly in Lean.